

24EC2211

Home Assignment - 1

1) Analyze the CV characteristic of MOS structure and nMOSFET?

Ans:

1) C-V characteristics of MOS structure (MOS capacitor):

- A MOS Capacitor Consist of Metal-Oxide-Semiconductor.
- Assume a p-type Substrate (standard for nmos).

The CV Curves plot Capacitance (C) Versus gate Voltage (V_G) three regions:

1) Accumulation: ($V_G < 0$)

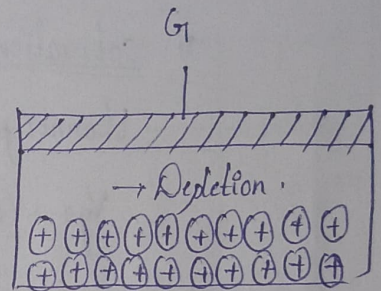
- Holes (majority carriers) accumulate at Si-SiO₂.
- Depletion Width 0
- Capacitance is maximum and approximately with oxide Capacitance.

$$C = C_{ox}$$

2) Depletion: ($0 < V_G < V_T$)

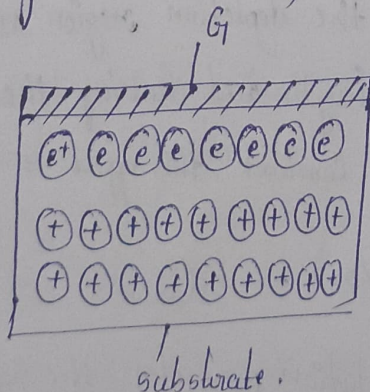
- Depletion region forms and widens with V_G
- Effective Capacitance decrease due to series Combination

$$C = \frac{C_{ox} C_{dep}}{C_{ox} + C_{dep}}$$



3) Inversion Mode ($V_G > V_T$)

- strong inversion electron forms at the surface



CV characteristics of nMOSFET $\rightarrow$

- An nMOSFET adds source and drain, so the gate capacitance is distributed.

$$C_G = C_{gs} + C_{gd} + C_{gb} \rightarrow (1)$$

operating region depends on

1) Cut off:

No channel behaviour to Mos Capacitor in depletion.

- $V_{gs} = 0, V_{ds} = \text{high Voltage}$

- $V_{gs} > V_t, V_{ds} = 0 \quad [\therefore I_D = 0]$

2) Non-saturation (ohmic region)

Continuous channel from source to drain

- $V_{gs} > V_t, V_{ds} < V_{gs} - V_t \quad [\therefore I_D = \text{min}]$

3) Saturation

channel pinches off near drain.

- $V_{gs} > V_t, V_{ds} \geq V_{gs} - V_t \quad [\therefore I_D = \text{max}]$

2) Detail the non-linear effect observed in MOSFET (channel length modulation & substrate effect).

non linear effect observed in MOSFET arises due to practical device behaviour. channel length modulation (CLM) occurs in saturation region when an increase in drain to source Voltage V_{ds} widens the depletion region near to drain there be reducing the effective channel length and causing the drain current to increase slightly instead of remaining constant. this effect is

$$I_D = I_{Dsat} (1 + \lambda V_{ds}) \rightarrow (1)$$

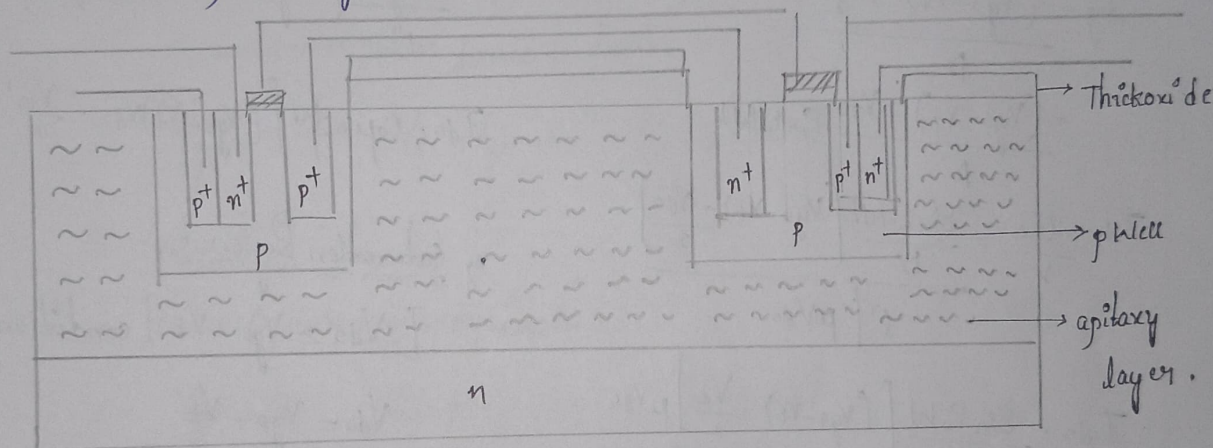
where λ is channel length modulation parameter, the substrate body effect occur when the source and substrate are different potential

$$V_T = V_{T0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F}).$$

3) Detail the CMOS twin-tube fabrication process?

The CMOS twin tube fabrication process is a technology in which NMOS & PMOS transistors are formed in two separate well (p-well & n-well) on the same silicon substrate. A lightly doped wafer is first taken and by using ion implantation and photolithography a p-well is created for NMOS and n-well is created for PMOS, allowing independent control of device parameter. After isolation using field oxide, a thin gate oxide is grown and polysilicon gate formed. Then n source and drain regions are implanted in the p-well of NMOS, and p region are implanted in the n-well for PMOS.

- To overcome latch up problem (one type of destructions of a chip) we are using two-well & epitaxial layer.



$$\text{Threshold Voltage } V_t = V_{T0} + \gamma (\sqrt{\phi_s + V_{DS}} - \sqrt{\phi_s})$$

4) Derive an expression of drain current of MOSFET and deduce the expression for deep triode, saturation, and saturation with channel length modulation?

Sol:

channel nmos :->

- Gradual channel approximation (GCA)
- Constant mobility μ_n
- Oxide capacitance per unit area C_{ox}
- Threshold Voltage V_t
- channel length L and Width W
- Gate Voltage V_{GS} & drain Voltage V_{DS} .

$$I_{ds} = -Q_{ds} = \frac{\text{charge induced in channel } (Q_c)}{\text{transist time } (T)}$$

$$\text{transist time } (T_{sd}) = \frac{\text{length of channel}}{\text{velocity}} \quad \therefore [\text{Velocity} = \mu_n E_{ds}] \quad \therefore \left[E_{ds} = \frac{V_{ds}}{L} \right]$$

$$T_{sd} = \frac{L^2}{\mu V_{ds}} \rightarrow (2)$$

$$I_{ds} = \frac{\text{charge induced } (Q_c)}{T_{sd}} = \frac{Q_c}{L^2 / \mu V_{ds}}$$

1) Non-saturation

$$\text{Where } V_{ds} = V_{gs} - V_t$$

$$Q_c = E_g \epsilon_{in} \epsilon_0 W L$$

$$E_g = \frac{(V_{gs} - V_t) - \frac{V_{ds}}{2}}{D}$$

$$= \frac{(V_{gs} - V_t) - \frac{V_{ds}}{2}}{D} \epsilon_{in} \epsilon_0 W L$$

$$I_{ds} = \frac{Q_c}{T} = \frac{\epsilon_{in} \epsilon_0 W L \left[(V_{gs} - V_t) - \frac{V_{ds}}{2} \right]}{L^2 / \mu V_{ds}}$$

$$I_{ds} = \frac{\epsilon_{in} \epsilon_0 W}{D} \left[(V_{gs} - V_t) - \frac{V_{ds}}{2} \right] \mu V_{ds}$$

$$= \frac{k W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$\boxed{I_{ds} = \beta \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]}$$

2) Gate channel Capacitance

$$C_g = \frac{\epsilon_{in} \epsilon_0 W L}{D} \quad k = \frac{C_g \mu}{W L}$$

$$I_{ds} = \frac{C_g \mu}{L^2} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$I_{ds} = C_g \mu \frac{W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

3) Saturation Region : $\rightarrow$

$$[V_{ds} \geq V_{gs} - V_t] \text{ condition}$$

$$V_{ds} = V_{gs} - V_t$$

from non-saturation equation

$$I_{ds} = \frac{k W}{L} \left[\frac{(V_{gs} - V_t)^2}{2} \right]$$

$$I_{ds} = \frac{\beta}{2} (V_{gs} - V_t)^2$$

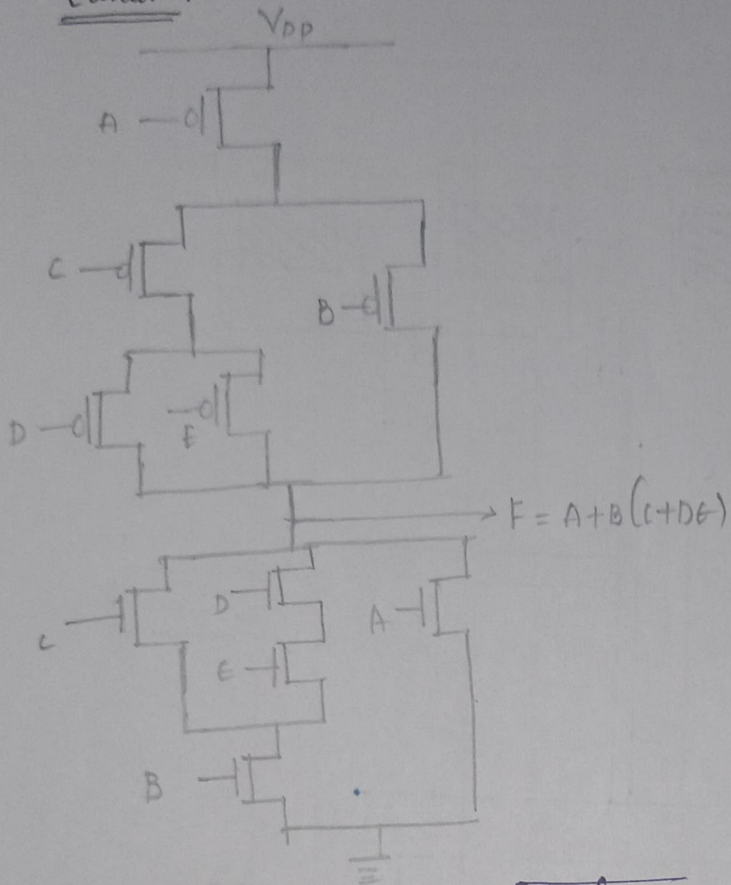
$$= \frac{e \mu}{2 L^2} (V_{gs} - V_t)^2$$

$$\boxed{I_{ds} = C_g \mu \frac{W}{2 L} (V_{gs} - V_t)^2}$$

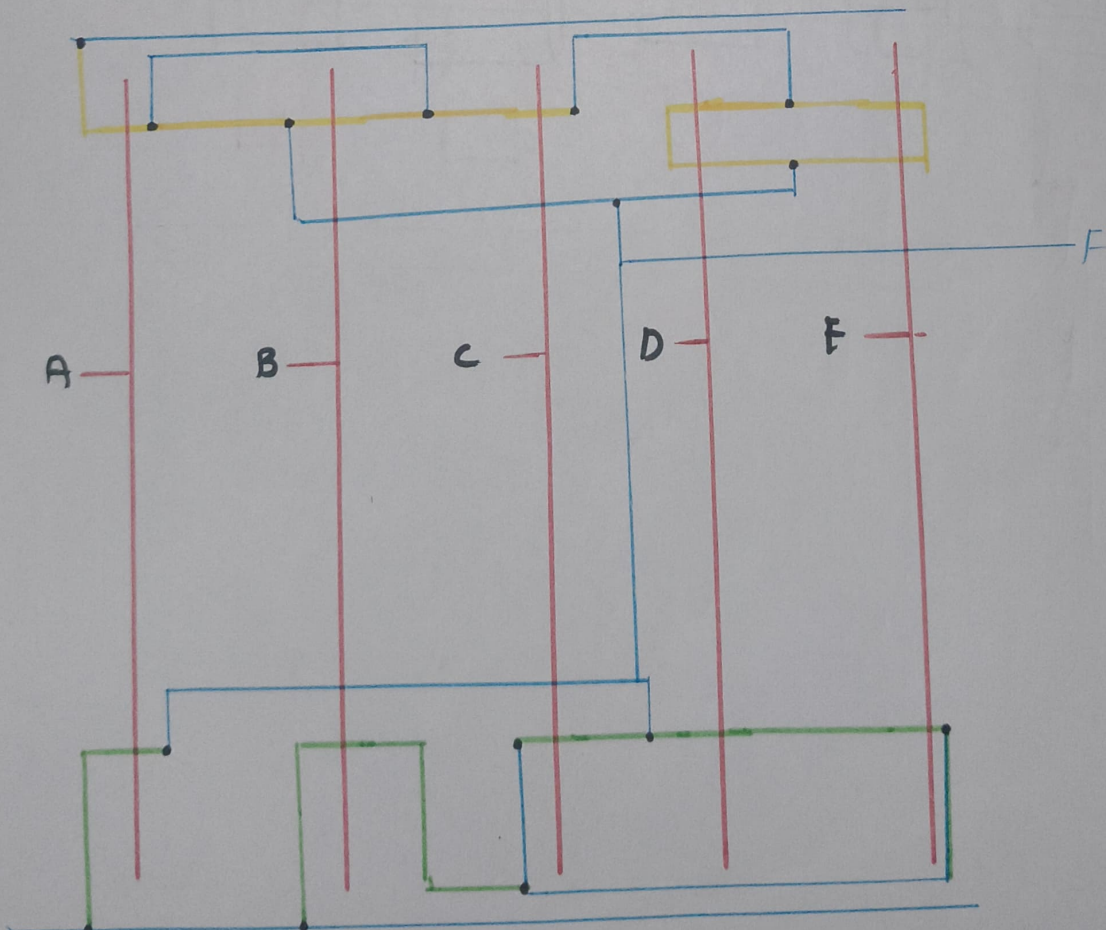
5)

Draw the stick Diagram and layout of function $F = A + B(C + DE)$!

Circuit:



→ stick Diagram for given $F = A + B(C + DE)$



Layout Diagram!→

